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On Coordinated Control of Multiple Flexible Joint Robots Holding a Constrained Object †

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Abstract

Coordinated control of multiple flexible joint robots holding a constrained object is addressed in this paper. A combined adaptive and robust control scheme, consisting of an adaptive coordinated controller and a robust joint torque controller, is proposed to control simultaneously the motion of the object, the internal force among the robots, the contact force between the object and environment, and the joint torques. The proposed control scheme is developed assuming that all system parameters, including the joint flexibility values, are unknown except for some of their bounds. Stability analysis of the control system shows that with the proposed control scheme, asymptotic motion and force tracking is achieved. Simulation results illustrate the effectiveness of the proposed control method.

1. Introduction

There is growing interest in the development of coordinated multi-manipulator systems. The potential applications of such systems cover a wide range, such as, lifting a heavy object or assembling mating parts, and all other tasks recognized to be beyond the capability of a single manipulator. In all of these applications, multiple robot manipulators should be controlled in a coordinated fashion to handle complicated and dexterous tasks skillfully.

The control issues of multi-manipulator systems are very complex. When multiple robot manipulators hold a common object in cooperative manipulation, closed kinematic chains are formed. These impose kinematic and dynamic constraints and result in internal forces (stress, i.e. pushing and pulling forces) on the object and robots. If the manipulators are not carefully coordinated, conflicting motions among them may cause excessive stress or even damages to the object or robots. The control issue is further complicated if the manipulated object is required to be in contact with the environment. Force arising from contact with the environment has a significant impact on the object and robotic system, and should be controlled simultaneously with the motion as well as internal forces. These problems have been investigated by many researchers. Numerous control methods have been proposed for the control of coordinated robots. These methods can be classified into two categories: i) master/slave control; and ii) hybrid position/force control. In the master/slave control schemes [1, 2], one or a group of arms play the role of the master, and the rest of the arms form the slave group which moves in conjunction with the master. Once the trajectory of the master is planned, the trajectory of the slave is derived from a set of holonomic constraints. In the hybrid position/force control approaches [3, 4, 5], the object motion and force trajectories are specified based on the overall dynamics of the multi-manipulator system, and controlled by two explicit control laws, namely, motion control and force control laws. To overcome the effects of uncertainties in the system dynamics, several adaptive and robust control schemes for coordinated robots have been proposed [7, 8, 9].

In the above mentioned control schemes, the robot manipulators are assumed to be rigid. The work [10, 12] has pointed out that control of robots based on rigid-body dynamics formulation is inadequate in dealing with stringent operating conditions. The elasticity of the transmission elements between actuators and links has a significant influence on the robot dynamics. In some cases, joint flexibility can even lead to instability if it is neglected in the control design. Joint flexibility causes difficulties in modeling and control. Much effort has been devoted to the control of flexible joint robots. A number of control strategies based on different techniques have been proposed for the control of single flexible joint robot [11, 12, 13, 14]. But, only a few studies, so far, have addressed the coordinated control of multiple flexible joint robots. Hu and Goldenberg [16] developed a control scheme for coordinated robot arms in the presence of joint flexibility. Asymptotic motion and force tracking is achieved. Jankowski etc. [17] extended their nonlinear decoupling control law to multiple flexible joint robots. It was shown that the closed system has high-precision tracking capabilities. Shaheen [18] proposed a control method for multiple flexible joint robots by utilizing the constrained motion control technique. Their results rely on the full knowledge of the system dynamics.

In this paper, adaptive and robust control of coordinated multiple flexible joint robot manipulators is considered. Concise dynamic models of multiple flexible joint robot manipulators holding a constrained object are derived. Based on the two-stage control strategy [15], a control scheme, consisting of an adaptive coordinated controller and a robust joint torque controller, is proposed to control the motion, internal force, contact force and joint torque simultaneously. The combined adaptive and robust control scheme is developed under the assumption that all system parameters, including the joint flexibility values, are unknown except for some of their bounds. Stability analysis

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of the control system shows that with the proposed control schemes, asymptotic motion, force and joint torque trackings are achieved. Simulation results illustrate the effectiveness of the proposed control methods.

2. Dynamic Formulation of Multiple Flexible Joint Robot System

Consider γ robot arms holding a rigid object moving along a rigid surface in a coordinated fashion. To facilitate the dynamic formulation, the following assumptions are made.

- Each manipulator is nonredundant, and all manipulators have the same number of joints, say n.
- All end-effectors are firmly holding the object at γ specified contact points, so that there is no relative motion at the contact points and there is perfect force transmission between the robots and object.

2.1 Dynamic Equation of Multiple Flexible Joint Robots

The dynamic equations of multiple flexible joint robots with joint torque measurements can be described as follows [12]:

$$D_j(q_{l,j})\ddot{q}_{l,j} + C_j(q_{l,j},\dot{q}_{l,j})\dot{q}_{l,j} + G_j(q_{l,j}) = \tau_{s,j} - J_j^T F_j \quad (1)$$

$$I_{m,j}\ddot{q}_{m,j} + B_{m,j}\dot{q}_{m,j} + \tau_{s,j} = u_j \quad (2)$$

and

$$\tau_{s,j} = K_{s,j}(q_{m,j} - q_{l,j}) \quad (3)$$

where the subscript j denotes the robot manipulator number, $j = 1, 2, ..., \gamma$. D_j is the $n \times n$ positive definite, symmetric inertia matrix of robot j; $C_j(q_l, \dot{q}_l)$ is the $n \times n$ matrix containing Coriolis, centrifugal terms; $G_j(q_l)$ is the $n \times 1$ vector of gravitational forces; $I_{m,j}$ is the $n \times n$ constant diagonal matrix representing the inertia of the rotors and gears; $B_{m,j}$ is the $n \times n$ diagonal matrix of damping terms and $K_{s,j}$ is a $n \times n$ diagonal matrix of the joint torsional stiffness. $q_{l,j}$ and $q_{m,j}$ are n dimensional vectors, represent the link angles and rotor angles respectively. u_j is the input torque control vector, $\tau_{s,j}$ is the vector of joint torque measurements and F_j is the vector of end-effector force of robot j. J_j is the $n \times n$ Jacobian matrix of robot j from joint space to the end-effector frame.

For simplicity, let

$$q_l = [q_{l,1}^T, ..., q_{l,\gamma}^T]^T \in R^{n\gamma}$$
$$q_m = [q_{m,1}^T, ..., q_{m,\gamma}^T]^T \in R^{n\gamma}$$
$$\tau_s = [\tau_{s,1}^T, ..., \tau_{s,\gamma}^T]^T \in R^{n\gamma}$$
$$u = [u_1^T, ..., u_\gamma^T]^T \in R^{n\gamma}$$
$$F = [F_1^T, ..., F_\gamma^T]^T \in R^{n\gamma}$$

Utilizing block diagonal metrices, the dynamic equations (1)–(3) can be expressed concisely as:

$$D(q_l)\ddot{q}_l + C(q_l,\dot{q}_l)\dot{q}_l + G(q_l) = \tau_s - J^T F \quad (4)$$

$$I_m\ddot{q}_m + B_m\dot{q}_m + \tau_s = u \quad (5)$$

and

$$\tau_s = K_s(q_m - q_l) \quad (6)$$

With joint torque measurements τ_s, from equation (6), we have

$$q_m = K_s^{-1}\tau_s + q_l$$

Substituting this equality into equation (5), we get

$$I_{ts}\ddot{\tau}_s + B_{ts}\dot{\tau}_s + h_{ts}(r,\dot{r},\ddot{r},\tau_s) = u \quad (7)$$

where

$$I_{ts} = I_m K_s^{-1}$$
$$B_{ts} = B_m K_s^{-1}$$
$$h_{ts} = I_m\ddot{q}_l + B_m\dot{q}_l + \tau_s$$

2.2 Dynamic Equation of the Constrained Object

The dynamic equation for the common object in constrained motion is given as [7]:

$$M_o(X_o)\ddot{X}_o + N_o(X_o,\dot{X}_o)\dot{X}_o + G_o(X_o) = J_o^T F - F_c \quad (8)$$

where M_o is the inertia matrix of the object, N_o is the Coriolis, centrifugal terms; G_o is the vector of gravitational forces. X_o is a n dimensional vector, denotes the Cartesian coordinate of the mass center of the object. J_{oj} is the $n \times n$ Jacobian matrix from the object frame to the jth end-effector frame, $J_o = [J_{o1}^T, ..., J_{o\gamma}^T]^T \in R^{n\gamma \times n}$, and F_c is the constraint force exerted by the environment on the object.

The environmental constraint is described as a holonomic smooth manifold. The constraint can be expressed with X_o as

$$\Psi(X_o) = 0 \quad (9)$$

where the mapping Ψ: $R^n \rightarrow R^m$ $n > m$ is twice continuously differentiable. The Jacobian matrix of the above constraint equation is

$$J_c(X_o) = \frac{\partial\Psi(X_o)}{\partial X_o} \quad (10)$$

Since $\Psi(X_o) = 0$ is identically satisfied, it is evident that $J_c(X_o)\dot{X}_o = 0$. When the object is moving along the constrained surface, the constraint force on the object is given by [19]

$$F_c = J_c^T(X_o)\lambda \quad (11)$$

where $\lambda \in R^m$ is the associated Lagrangian multiplier.

2.3 The Combined Robots/Object Dynamics

Since each robot manipulator holds the object firmly at the contact point thus forming a close kinematic chain, we have the following relationship

$$J\dot{q}_l = J_o\dot{X}_o \quad (12)$$

Assuming that all robot manipulators work in non-singular work space. J is then a non-singular matrix, we have

$$\dot{q}_l = J^{-1}J_o\dot{X}_o \quad (13)$$

$$\ddot{q}_l = J^{-1}J_o\ddot{X}_o + \frac{d}{dt}(J^{-1}J_o)\dot{X}_o \quad (14)$$
$$= J^{-1}J_o\ddot{X}_o + (J^{-1}\dot{J}_o - J^{-1}\dot{J}J^{-1}J_o)\dot{X}_o$$

As we known, the matrix J_o is full rank but not square. Thus, for a given trajectory $X_o(t)$, the applied force F is not unique and can be expressed in the form of

$$F = J_o^+\{M_o(X_o)\ddot{X}_o + N_o(X_o,\dot{X}_o)\dot{X}_o + G_o(X_o) + F_c\} + F_I \quad (15)$$

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where $J_o^+ = J_o(J_o^T J_o)^{-1}$. F_I represents an internal force, that produces zero generalized force and does not affect the motion of the object. It can be any vector in the null space of J_o^T. The values of F_I are usually specified according to desired squeezing force on the object [6] and load distribution among the robot arms [4, 7].

With the above relations, substituting equation (13), (14) and (15) into (4), we get

$$M_l\ddot{X}_o + N_l\dot{X}_o + G_l = \tau_s - J^T J_o^+ F_c - J^T F_I \qquad (16)$$

with

$$M_l = (DJ^{-1}J_o + J^T J_o^+ M_o)$$
$$N_l = D\frac{d}{dt}(J^{-1}J_o) + CJ^{-1}J_o + J^T J_o^+ N_o$$
$$G_l = G + J^T J_o^+ G_o$$

Multiplying both sides from left by $J_o^T J^{-T}$ and noticing that $J_o^T J^{-T} J^T F_I = J_o^T F_I = 0$, we have

$$M\ddot{X}_o + N\dot{X}_o + W = J_o^T J^{-T}\tau_s - F_c \qquad (17)$$

where

$$M = J_o^T J^{-T} D J^{-1} J_o + M_o$$
$$N = J_o^T J^{-T} D\frac{d}{dt}(J^{-1}J_o) + J_o^T J^{-T} C J^{-1} J_o + N_o$$
$$W = J_o^T J^{-T} G + G_o$$

2.4 Reduced Dynamic Model

Since the presence of m constraints causes the multi-robot system to lose m degrees of freedom, $n-m$ linearly independent coordinates are sufficient to characterize the constrained motion of the object. Let us partition the vector X_o as

$$X_o = \begin{bmatrix} X_o^1 \\ X_o^2 \end{bmatrix} \qquad (18)$$

with

$$X_o^1 = [X_{o1}^1, \cdots X_{o(n-m)}^1]^T \in R^{n-m}$$
$$X_o^2 = [X_{o1}^2, \cdots X_{om}^2]^T \in R^m$$

and according to the implicit function theory, from the constraint equation (9), X_o^2 can be expressed as $X_o^2 = \sigma(X_o^1)$, [19].

Defining

$$L(X_o^1) = \begin{bmatrix} I_{n-m} \\ \frac{\partial\sigma(X_o^1)}{\partial X_o^1} \end{bmatrix} \qquad (19)$$

we have

$$\dot{X}_o = \begin{bmatrix} \dot{X}_o^1 \\ \dot{X}_o^2 \end{bmatrix} = L(X_o^1)\dot{X}_o^1$$
$$\ddot{X}_o = L(X_o^1)\ddot{X}_o^1 + \dot{L}(X_o^1)\dot{X}_o^1$$

Substituting above relations into (16) and (17), we get

$$M_l L(X_o^1)\ddot{X}_o^1 + H_l\dot{X}_o^1 + G_l = \tau_s - J^T J_o^+ F_c - J^T F_I \qquad (20)$$

$$ML(X_o^1)\ddot{X}_o^1 + B(X_o^1, \dot{X}_o^1)\dot{X}_o^1 + W(X_o^1) = J_o^T J^{-T}\tau_s - J_c^T\lambda \qquad (21)$$

where

$$H_l(X_o^1, \dot{X}_o^1) = M_l\dot{L} + N_l L$$

$$B(X_o^1, \dot{X}_o^1) = M\dot{L}(X_o^1) + N(X_o^1, \dot{X}_o^1)L(X_o^1) = J_o^T J^{-T} H_l(X_o^1, \dot{X}_o^1)$$

It can be shown that the dynamic equations have the following properties:

Property 1. The LHS of equation (20) and (21) can be expressed as

$$M_l L(X_o^1)\ddot{X}_o^1 + H_l\dot{X}_o^1 + G_l = Y_l^1(X_o^1, \dot{X}_o^1, \ddot{X}_o^1)P \qquad (22)$$

$$ML(X_o^1)\ddot{X}_o^1 + B(X_o^1, \dot{X}_o^1)\dot{X}_o^1 + W(X_o^1) = Y^1(X_o^1, \dot{X}_o^1, \ddot{X}_o^1)P \qquad (23)$$

where $Y^1(.)$ is an $n \times r$ matrix of known functions, referred to as the regressor, and P is an r-dimensional vector of parameters.

Property 2. The matrix M is a bounded positive definite, symmetric matrix, and

$$\dot{M} - 2N = J_o^T J^{-T}\dot{D}J^{-1}J_o + \dot{M}_o + 2(J_o^T J^{-T})D\frac{d}{dt}(J^{-1}J_o)$$
$$-2(J_o^T J^{-T})D\frac{d}{dt}(J^{-1}J_o) - 2J_o^T J^{-T} C J^{-1} J_o - 2N_o$$
$$= J_o^T J^{-T}(\dot{D} - 2C)J^{-1}J_o + (M_o - 2N_o) \qquad (24)$$

is skew symmetric. Let $A(X_o^1) = L^T(X_o^1)ML(X_o^1)$, then

$$\dot{A}(X_o^1) - 2L^T(X_o^1)B(X_o^1, \dot{X}_o^1) = L^T(\dot{M} - 2N)L \qquad (25)$$

is also skew symmetric.

Property 3. Since $J_c\dot{X}_o = 0$, i.e. $J_c(X_o^1)L(X_o^1)\dot{X}_o^1 = 0$, and X_o^1 is linearly independent, we have

$$J_c L(X_o^1) = 0, \qquad L^T(X_o^1)J_c^T = 0$$

Thus, multiplying equation (21) from left by L^T, yields

$$A(X_o^1)\ddot{X}_o^1 + L^T B(X_o^1, \dot{X}_o^1)\dot{X}_o^1 + L^T W(X_o^1) = L^T J_o^T J^{-T}\tau_s \qquad (26)$$

With the above derived dynamic equations and properties, we are now ready to design the adaptive and robust control laws.

3. Adaptive and Robust Control Scheme

Coordinated manipulation of multiple robots moving an object along and in contact with a rigid surface requires simultaneous control of both motion and contact force, as well as precise internal force regulation. The objective of flexible joint robots control is to determine the input torque to meet these requirements. Based on the two-stage control strategy [15], a control scheme consisting of a coordinated controller and a joint torque controller is developed in such a way that the coordinated controller is designed to generate the *"desired joint torque"* τ_{sd}, and the joint torque controller computes the required control torque u so that joint torque τ_s can follow the desired joint torque τ_{sd}, and thus the whole system achieves the control objective.

In practice, it is usually impossible to obtain the exact dynamic model of a multiple flexible joint robots system. To deal with the parameter uncertainty in the multiple robot system, adaptive and robust controllers are developed assuming that all system parameters including the joint flexibility values are unknown except for some of their bounds.

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3.1 Adaptive Coordinated Controller

Let $X_d(t)$, λ_d and $F_{Id}(t)$ denote the desired motion of the object, the desired contact force and the desired internal force respectively, and define the tracking errors:

$$e_x = X_o^1 - X_d^1$$
$$e_f = \lambda - \lambda_d$$
$$e_{If} = F_I - F_{Id}$$
$$e_t = \tau_s - \tau_{sd}$$

and the auxiliary signals:

$$v_l^1 = \dot{X}_d^1 - \Lambda_l(X_o^1 - X_d^1)$$
$$r_l = \dot{X}_o^1 - v_l^1 = (\dot{X}_o^1 - \dot{X}_d^1) + \Lambda_l(X_o^1 - X_d^1)$$

The proposed coordinated controller has three components, namely, a motion controller τ_{mc}, a contact force controller τ_{fc} and a internal force controller τ_{If}.

$$\tau_{mc} = \hat{M}_l L\dot{v}_l^1 + \hat{H}_l v_l^1 + \hat{G}_l - K_{lD}J^{-1}J_o L r_l \quad (27)$$

$$\tau_{fc} = J^T J_o^+[J_c^T\lambda_d + J_c^T k_{fP}(\lambda_d - \lambda) + J_c^T k_{fI}\int_0^t(\lambda_d - \lambda)dt] \quad (28)$$

$$\tau_{If} = J^T[F_{Id} + K_{Ip}(F_{Id} - F_I) + K_{II}\int_0^t(F_{Id} - F_I)dt] \quad (29)$$

where K_{lD}, k_{fP}, k_{fI}, K_{IP}, K_{ID} and Λ_l are diagonal matrices of positive gains.

The desired joint torque τ_{sd} is thus generated as

$$\tau_{sd} = \tau_{mc} + \tau_{fc} + \tau_{If} \quad (30)$$

Substituting (30) into (20) and re-arranging the terms leads to the following equations:

$$M_l L\dot{r}_l + H_l r_l + K_{lD}J^{-1}J_o L r_l = Y_l^1(X_o^1, \dot{X}_o^1, v_l^1, \dot{v}_l^1)\tilde{P} + e_t - J^T J_o^+[J_c^T(I + k_{fP})e_f + J_c^T k_{fI}\int_0^t e_f dt] - J^T[(I + K_{IP})e_{If} + K_{II}\int_0^t e_{If}dt] \quad (31)$$

$$ML\dot{r}_l + Br_l + J_o^T J^{-T}K_{lD}J^{-1}J_o L r_l = Y^1(X_o^1, \dot{X}_o^1, v_l^1, \dot{v}_l^1)\tilde{P} + J_o^T J^{-T}e_t - J_c^T[(I + k_{fP})e_f + k_{fI}\int_0^t e_f dt], \quad (32)$$

and

$$A\dot{r}_l + L^T B r_l + L^T J_o^T J^{-T}K_{lD}J^{-1}J_o L r_l = L^T Y^1(X_o^1, \dot{X}_o^1, v_l^1, \dot{v}_l^1)\tilde{P} + L^T J_o^T J^{-T}e_t \quad (33)$$

with $\tilde{P} = \hat{P} - P$, and the parameter update law as

$$\dot{\hat{P}} = -\Gamma^{-1}Y^{1T}Lr_l = -\Gamma^{-1}Y_l^{1T}(J^{-1}J_o)Lr_l \quad (34)$$

3.2 Robust Joint Torque Controller

Joint torque controller is designed to drive the actual joint torque to follow the desired joint torque τ_{sd}. To deal with the uncertainty in the rotor subsystem, sliding mode control approach is used. We define the sliding surface as

$$s_t = \dot{\tau}_s - \dot{\tau}_{sd} + \Lambda_t(\tau_s - \tau_{sd}) = \dot{e}_t + \Lambda_t e_t \quad (35)$$

and denote

$$v_t = \dot{\tau}_s - s_t = \dot{\tau}_{sd} - \Lambda_t e_t$$

For tracking the desired joint torque τ_{sd}, the joint torque sliding control law is given as [15]:

$$u = \hat{I}_{ts}\dot{v}_t + \hat{B}_{ts}v_t + \hat{h}_{ts} - K_{tD}s_t - k_t sgn(s_t) \quad (36)$$

where $\hat{I}_{ts}$, $\hat{B}_{ts}$ and $\hat{h}_{ts}$ are the estimated nominal parameters. K_{tD} and Λ_t are diagonal positive definite gain matrices, k_t is a positive definite gain matrix with minimum eigenvalue $\lambda_{k_t}^{min}$.

Substituting this control law into (7), yields:

$$I_{ts}\dot{s}_t + B_{ts}s_t + K_{tD}s_t = w(\dot{r}, \ddot{r}, \tau_s, v_t, \dot{v}_t) - k_t sgn(s_t) \quad (37)$$

where $w = (\hat{I}_{ts} - I_{ts})\dot{v}_t + (\hat{B}_{ts} - B_{ts})v_t + (\hat{h}_{ts} - h_{ts})$ represents the dynamic uncertainty in the rotor subsystem. As the uncertainty is bounded, it could be expressed as

$$\| w \| \leq \varrho \quad (38)$$

3.3 Stability Analysis

The stability and convergence properties of the entire control system are analyzed based on Lyapunov stability theory. With the proposed control scheme, we have the following theorem:

Theorem 1 *Consider the multiple flexible joint robot system holding a constrained object. The adaptive coordinated controller (30), with parameter update law (34), and joint torque sliding controller (36) stabilize the system and achieve global asymptotic convergence, i.e.*

$$\lim_{t\to\infty} e_x = 0, \quad \textit{and} \quad \lim_{t\to\infty} e_t = 0.$$

And if the system is persistently exciting, the contact force and internal force tracking are achieved with

$$\lim_{t\to\infty} e_f = 0, \textit{ and } \lim_{t\to\infty} e_f = 0.$$

Proof: Choosing the Lyapunov function candidate as

$$V = \frac{1}{2}r_l^T A r_l + \frac{1}{2}\tilde{P}^T\Gamma\tilde{P} + \frac{1}{2}s_t^T I_{ts}s_t + e_t^T\Lambda_t^T K_{tD}e_t > 0 \quad (39)$$

the differentiation of V is

$$\dot{V} = r_l^T A\dot{r}_l + \frac{1}{2}r_l^T\dot{A}r_l + \tilde{P}^T\Gamma\dot{\tilde{P}} + s_t^T I_{ts}\dot{s}_t + 2e_t^T\Lambda_t^T K_{tD}\dot{e}_t \quad (40)$$

Using the fact that $(\dot{A} - 2L^T B)$ is skew symmetric and equation (33),

$$r_l^T A\dot{r}_l + \frac{1}{2}r_l^T\dot{A}r_l = r_l^T(-L^T J_o^T J^{-T}K_{lD}J^{-1}J_o L r_l + L^T J_o^T J^{-T}e_t + L^T Y^1(X_o^1, \dot{X}_o^1, v_l^1, \dot{v}_l^1)\tilde{P}) \quad (41)$$

together with equation (37), we have

$$\dot{V} = -r_l^T L^T J_o^T J^{-T}K_{lD}J^{-1}J_o L r_l + r_l^T L^T J_o^T J^{-T}e_t - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t - e_t^T\Lambda_t^T K_{tD}\Lambda_t e_t + \tilde{P}^T(Y^{1T}Lr_l + \Gamma\dot{\tilde{P}}) + s_t^T(w - k_t sgn(s_t)) \quad (42)$$

With the parameter update law (34)

$$\dot{V} = -\begin{bmatrix} (J^{-1}J_o Lr_l)^T & e_t^T \end{bmatrix} Q \begin{bmatrix} J^{-1}J_o Lr_l \\ e_t \end{bmatrix} - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t + s_t^T(w - k_t sgn(s_t)) \quad (43)$$

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where

$$Q = \begin{bmatrix} K_{lD} & -\frac{1}{2}I \\ -\frac{1}{2}I & \Lambda_t^T K_{tD} \Lambda_t \end{bmatrix}$$

For matrix Q to be a positive definite matrix, the requirement is

$$\Lambda_t^T K_{tD} \Lambda_t > \frac{1}{4} K_{lD}^{-1}$$

Since K_{tD}, Λ_t and K_{lD} are diagonal matrices of controller gains, we can choose them properly to meet the requirement, such that $Q > 0$. The last term in (43)

$$\begin{aligned} s_t^T(w - k_t sgn(s_t)) &\leq (\| s_t^T w \| - \| s_t^T k_t \|) \\ &\leq \| s_t \| (\varrho - \lambda_{k_t}^{min}) \end{aligned} \tag{44}$$

To meet the requirement of the sliding condition, the gain k_t is selected such that the minimum eigenvalue $\lambda_{k_t}^{min}$ satisfies

$$\lambda_{k_t}^{min} > \varrho + \epsilon \tag{45}$$

Then

$$\dot{V} < -\begin{bmatrix} (J^{-1} J_o L r_l)^T & e_t^T \end{bmatrix} Q \begin{bmatrix} J^{-1} J_o L r_l \\ e_t \end{bmatrix} - s_t^T B_{ts} s_t$$
$$-\dot{e}_t^T K_{tD} \dot{e}_t - \epsilon \| s_t \| \tag{46}$$

The whole system is asymptotically stable in the sense of Lyapunov. We have

$$\lim_{t \to \infty} r_l = 0, \qquad \lim_{t \to \infty} e_t = 0.$$

And if the system is persistently exciting, the estimated parameters $\hat{P}$ converge asymptotically to the true values P, then from equation (32) and (31), we have

$$\lim_{t \to \infty} e_f = 0, \qquad \lim_{t \to \infty} e_{If} = 0$$

Usually, the estimated parameters may not converge to their true values as persistently exciting condition is not satisfied. But, the error of parameter estimation is bounded. From equation (32) and (31), it is clear that the force tracking errors are bounded too. The integral action and the proportional gain $(I + k_{fP})$ and $(I + K_{IP})$ in τ_{fc} and τ_{If} can reduce the errors and make them as small as required.

4. Simulation Examples

For illustration, the proposed control scheme is applied to coordinated manipulation of two planar robot manipulators. For simplicity, we assume that the two robots are identical, each has three joint-links, and the coordinate frames are located as shown in figure 1. The position of the object moving in a horizontal plane is determined by the coordinate x_o, y_o and the orientation θ_o. The system parameters are given in the Table 1.

Table 1. The parameter values of the system

$m_o = 3kg$	$r_1 = 0.25m$	$r_2 = 0.25m$
$m_1 = 1kg$	$l_1 = 1m$	$l_{c1} = 0.5m$
$m_2 = 1kg$	$l_2 = 1m$	$l_{c2} = 0.5m$
$m_3 = 0.25kg$	$l_3 = 0.25m$	$l_{c3} = 0.125m$
$I_{m1} = 0.02kgm^2$	$B_{m1} = 5.0\frac{Nm}{(rad/s)}$	$K_{s1} = 1000(Nm/rad)$
$I_{m2} = 0.015kgm^2$	$B_{m2} = 4.0\frac{Nm}{(rad/s)}$	$K_{s2} = 800(Nm/rad)$
$I_{m3} = 0.005kgm^2$	$B_{m3} = 2.0\frac{Nm}{(rad/s)}$	$K_{s3} = 600(Nm/rad)$

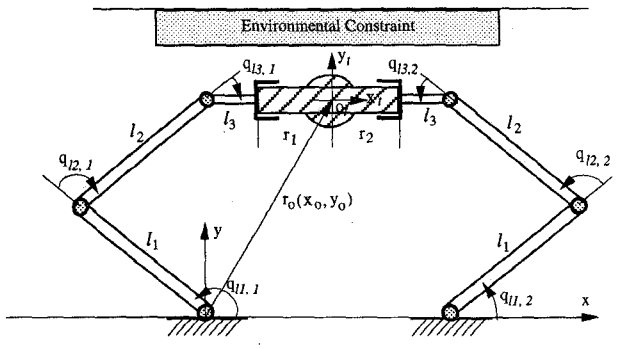


Figure 1: Coordinated manipulation of two robots holding a constrained object

Consider the situation in which the object is required to move along a straight line in contact with the constraint surface and exerts a constant force on the surface. Meanwhile, the robots are demanded to keep a constant "push" force (internal force) against each other.

The constraint is expressed in task space as

$$\Psi(X_o) = y_o - 1.0 = 0$$

and the constraint force is

$$F_c = [0 \;\; 1 \;\; 0]^T \lambda$$

The desired motion, internal force and contact force trajectories are assumed to be

$$\begin{aligned} x_d &= 0.5 + 0.25cos(t)(m) \\ \theta_d &= 0(rad) \\ F_{cd} &= [0 \;\; 1 \;\; 0]^T \lambda_d, \qquad \lambda_d = 15(N) \\ F_{Id} &= [25 \;\; 0 \;\; 0 \;\; -25 \;\; 0 \;\; 0]^T \end{aligned}$$

Simulations are performed using Simnon. The control scheme is digitally implemented with the sampling rate of 1msec. The motion, contact force and internal force tracking trajectories are plotted in Figure 2, 3 and 4, respectively. It is illustrated that the proposed control system has very good tracking performances. Simulation of the system neglecting the joint flexibility in the control design has also been performed. Vibration occurs in the response and the system is unstable. Due to the limit of the space, the result is omitted here.

5. Conclusions

In this paper, coordinated control of multiple flexible joint robots holding a constrained object is considered. A control scheme, consisting of an adaptive coordinated controller and a robust joint torque controller, is proposed to control the motion, internal force, contact force and joint torque simultaneously. The combined adaptive and robust control scheme is developed under the assumption that all system parameters, including the joint flexibility values, are unknown except for some of their bounds. Stability analysis of the control system shows that with the proposed control scheme, asymptotic motion, force and joint torque tracking is achieved. The proposed control scheme has the attractive feature that no restriction on joint flexibility, nor

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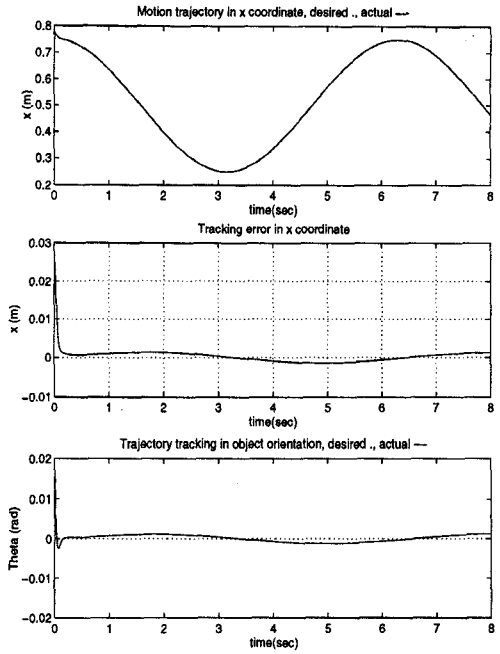


Figure 2: Motion tracking trajectory of the proposed control system

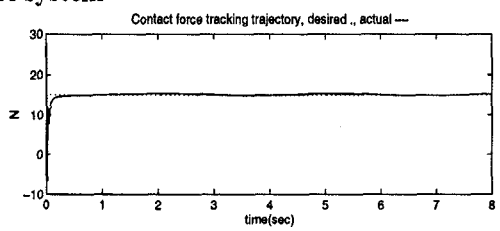


Figure 3: Contact force tracking trajectory of the proposed control system

exact knowledge of the dynamic parameters in the system is required. Compared with the previous work in the literature, the proposed control scheme provides a new approach to deal with the uncertainty in the multiple flexible joint robot system.

Simulation results illustrate the effectiveness of the proposed control methods.

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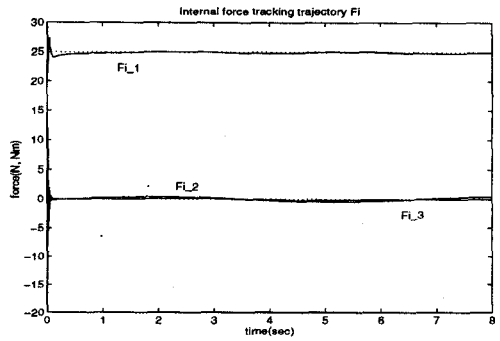


Figure 4: Internal force tracking trajectory of the proposed control system

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